

Normal-Model z -Scores and Percentiles

Computing z -scores, reading percentiles from a standard normal table, reversing the process to find a value, and comparing scores from different distributions

Directions

- Standardise with $z = \frac{x - \mu}{\sigma}$, and go back with $x = \mu + z\sigma$.
- Every proportion in the table below is the area to the LEFT of that z . For an area to the right, subtract from 1.
- Round z -scores to two decimal places and proportions to four, and state each answer as a proportion or a percentile as asked.

Standard normal table — use this for every problem on this sheet.

z	-2.00	-1.50	-1.28	-1.00	-0.50	0.00
Area left of z	0.0228	0.0668	0.1003	0.1587	0.3085	0.5000
z	0.50	0.67	1.00	1.28	1.50	2.00
Area left of z	0.6915	0.7486	0.8413	0.8997	0.9332	0.9772

1. Scores on a national test follow a normal model with mean $\mu = 500$ and standard deviation $\sigma = 100$. Find the z -score of a score of 640, and say in one sentence what that number means.

Answer: _____

2. For the same test, find the z -score of a score of 430. Say whether the score is above or below the mean, and by how many standard deviations.

Answer: _____

3. Use the table above. What proportion of a normal distribution lies below $z = 1.00$? Write that proportion, then state the matching percentile.

Answer: _____

4. Use the table. What proportion of a normal distribution lies below $z = 1.50$, and what proportion lies ABOVE it? Show the subtraction you used for the second answer.

Answer: _____

5. Adult heights in a population are modelled as normal with $\mu = 170$ cm and $\sigma = 8$ cm. Find the z -score of a height of 182 cm, then use the table to find the proportion of adults shorter than 182 cm.

Answer: _____

6. Using the same height model, find the proportion of adults whose height is between 162 cm and 182 cm. Standardise both endpoints first, then combine the two table entries.

Answer: _____

7. Still using $\mu = 170$ cm and $\sigma = 8$ cm: which height sits at the 90th percentile? Use the table to find the closest z -score to that percentile, then convert it back to a height.

Answer: _____

8. On the national test ($\mu = 500$, $\sigma = 100$), a scholarship goes to the top 2.28% of candidates. Use the table to find the z -score that cuts off that top group, then find the minimum score a candidate needs.

Answer: _____

9. Ana scored 640 on the national test ($\mu = 500$, $\sigma = 100$) and 28 on a different exam whose scores are normal with $\mu = 22$ and $\sigma = 4$. Compute both z -scores, compare them, and say which result is the relatively stronger performance and why.

Answer: _____

10. Of every 10,000 candidates who sit the national test, about 228 score above 700. If one candidate is chosen at random, what is the probability that they scored above 700? Give the answer as a fraction in lowest terms.

Answer: _____

11. A coach records the fitness-test scores of a squad of seven athletes: 12, 14, 16, 18, 20, 22, 24. Treat these seven as the whole population.

(a) Find the mean μ . (b) Find the standard deviation σ . (c) Find the z -score of the athlete who scored 24.

Answer: _____

12. A machine fills bottles. The fill volume is modelled as normal with $\mu = 500$ mL and $\sigma = 2.5$ mL, and the label promises at least 495 mL.

- (a) Find the z -score of 495 mL.
- (b) Use the table to find the proportion of bottles that are underfilled.
- (c) The plant will accept an underfilled proportion of at most 0.1003. Use the table to find the z -score with that area to its left, then find the mean μ the machine should be set to so that 495 mL sits exactly at that z , keeping $\sigma = 2.5$ mL.
- (d) State the one assumption every percentage above depends on, and what would go wrong without it.

Answer: _____